

ASSIGNMENT

Problem Identification:

Machine Learning- Supervised Learning – Classification

No. of rows and columns:

399 rows × 25 columns

Preprocessing method used :

Nominal data -> One Hot Encoding

It is used for to convert categorical value to numerical value.

Decision Tree:

	precision	recall	f1-score	support
0	0.90	1.00	0.95	45
1	1.00	0.93	0.97	75
accuracy			0.96	120
macro avg	0.95	0.97	0.96	120
weighted avg	0.96	0.96	0.96	120

Random forest:

	precision	recall	f1-score	support
0	0.98	1.00	0.99	45
1	1.00	0.99	0.99	75
accuracy			0.99	120
macro avg	0.99	0.99	0.99	120
weighted avg	0.99	0.99	0.99	120

SVM:

	precision	recall	f1-score	support
0	0.96	1.00	0.98	45
1	1.00	0.97	0.99	75
accuracy			0.98	120
macro avg	0.98	0.99	0.98	120
weighted avg	0.98	0.98	0.98	120

KNN:

	precision	recall	f1-score	support
0	0.88	1.00	0.94	45
1	1.00	0.92	0.96	75
accuracy			0.95	120
macro avg	0.94	0.96	0.95	120
weighted avg	0.96	0.95	0.95	120

Logistic Regression:

	precision	recall	f1-score	support
0	0.98	1.00	0.99	45
1	1.00	0.99	0.99	75
accuracy			0.99	120
macro avg	0.99	0.99	0.99	120
weighted avg	0.99	0.99	0.99	120

Bernoulli's NB:

	precision	recall	f1-score	support
0	0.94	1.00	0.97	45
1	1.00	0.96	0.98	75
accuracy			0.97	120
macro avg	0.97	0.98	0.97	120
weighted avg	0.98	0.97	0.98	120

Gaussian NB:

	precision	recall	f1-score	support
0	0.94	1.00	0.97	45
1	1.00	0.96	0.98	75
accuracy			0.97	120
macro avg	0.97	0.98	0.97	120
weighted avg	0.98	0.97	0.98	120

Multinomial NB:

	precision	recall	f1-score	support
0	0.67	0.98	0.79	45
1	0.98	0.71	0.82	75
accuracy			0.81	120
macro avg	0.82	0.84	0.81	120
weighted avg	0.86	0.81	0.81	120

Categorical NB:

	precision	recall	f1-score	support
0	0.99	1.00	1.00	150
1	1.00	1.00	1.00	249
accuracy			1.00	399
macro avg	1.00	1.00	1.00	399
weighted avg	1.00	1.00	1.00	399

Complement NB:

	precision	recall	f1-score	support
0	0.67	0.98	0.79	45
1	0.98	0.71	0.82	75
accuracy			0.81	120
macro avg	0.82	0.84	0.81	120
weighted avg	0.86	0.81	0.81	120

Final Model used here For Deployment is:

Logistic Regression with Accuracy: 99%

F1 Score value :

```
f1_macro value for {'penalty': 'l2', 'solver': 'lbfgs'}: 0.9916844900066377
```

ROC_AUC score = 100%

Comparing to all the models this models Accuracy and precision seems to be high. Thatswhy this

LogisticRegression model is choosen as the final model.